UNIT 2: SHAPES AND REGIONS

SESSION 4: BOUNDARY TRACKING PROCEDURES

SLO 2: ANALYZE AND REPRESENT OBJECT SHAPES USING BOUNDARY COORDINATES FOR FURTHER SHAPE-BASED ANALYSIS.

Coordinate Exploration

Observe the binary image grid below.

Assume the top-left pixel is (0,0).

0 1 2 3 4

0 0 0 0 0 0

1 0 1 1 1 0

2 0 1 1 1 0

3 0 1 1 1 0

4 0 0 0 0 0

Tasks

1. **Write the coordinates of any four boundary pixels.**

(1,1), (1,2), (1,3), (2,1)

2. **Identify one interior pixel of the object.**

(2,2)

Pattern Recognition

Two objects are represented by their boundary coordinates.

Object A

(1,1), (1,2), (1,3), (2,3), (3,3), (3,2), (3,1), (2,1)

Object B

(1,2), (2,3), (3,2), (2,1)

Questions

1. **Which object likely represents a square shape?**

Ans- Object A

2. **Which object likely represents a diamond or rotated shape?**

Ans- Object B

3. **Explain how boundary coordinates help identify shapes.**

Ans- Boundary coordinates help identify shapes by representing the position of the pixels along an object's outer boundary. By studying the arrangement, distances, and pattern of these coordinates, we can determine the shape and distinguish between different objects.